AI tutor evidence map

My role Math content and evaluation design

I mapped the evidence an AI tutor can use across grades 3–9, including patterns that signal a misconception. The map gives reviewers a concrete boundary for what the tutor can infer from student work.

Purpose and boundary

This document defines the productive and problematic evidence targets for Map 35 across Grades 3-9. It is an evidence-extraction reference, not the deterministic belief-state model itself.

Core principle: The LLM identifies student evidence and its support context. The deterministic model decides how that evidence updates facet beliefs.

Cross-grade progression

Grade band	How the evidence becomes more demanding
Grades 3-4	Relational meaning of equality; open sentences; unknowns; representing simple situations; keeping compared quantities on opposite sides.
Grades 5-6	Variables as quantities; substitution; one-step equations; multiplicative relationships; transition from arithmetic reasoning to generalized relationships.
Grades 7-8	Multi-step equations and inequalities; full solution sets; boundary inclusion; inequality direction; algebraic equivalence and symbolic modeling.
Grade 9	Literal equations, linear relations, and simple systems, while retaining the same foundational equality, variable, inverse-operation, and balance facets.

Progression rule: The facets do not reset at each grade. Later items combine earlier facets in longer, less familiar, or more abstract situations.